



Accelerated Tire Endurance

1 Scope

Note: Nothing in the standard supersedes applicable laws and regulations unless specific exemption has been obtained.

Note: In the event of conflict between the English and domestic language, the English language shall take precedence.

1.1 Purpose. To determine the durability of tires operating continuously at rated load conditions.

1.2 Foreword. The Accelerated Tire Endurance (ATE) test is a tire component test, run on a road schedule including highway and/or test track and gravel unpaved road types. This tire endurance (fatigue) test is accelerated by operating at the maximum tire load capacity. The time to complete the test is minimized by accumulating the required test distance of 72 000 km at a rapid rate of approximately 9650 km/week.

1.3 Applicability. This procedure applies to all tires used on GM passenger and light truck vehicles.

2 References

Note: Only the latest approved standards are applicable unless otherwise specified.

2.1 External Standards/Specifications.

ASTM F421 ASTM F1016

2.2 GM Standards/Specifications.

None

2.3 Additional References.

Tire and Rim Association (T&RA) Yearbook
European Tire (and) Rim Technical Organization (ETRTO) Yearbook

3 Resources

3.1 Facilities. Not applicable.

3.1.1 Test track or roadway suitable for continuous vehicle operation at 105 to 113 km/h and approved by General Motors (GM).

3.1.2 Secondary and gravel roads meeting the schedule requirements and approved by GM.

3.2 Equipment.

3.2.1 Tachograph or similar device

3.2.2 Ballast weight

3.2.3 Photographic equipment

3.2.4 Tire tread depth gauges

3.2.5 Tire inflation pressure gauges

3.2.6 Tire Pressure Monitor System (TPMS)

Note: These systems must be used for every GM ATE test. Any test where TPMS is not feasible, the exclusion of the TPMS must be approved by GM.

3.2.7 Four-wheel vehicle-alignment machine

3.2.8 Vehicle load scales

3.3 Test Vehicle.

3.3.1 A passenger car, light truck, or medium-duty truck vehicle with adequate load carrying capacity at each test position. Light trucks, or medium-duty trucks are used exclusively for light truck or commercial type tires. A light truck may also be used for passenger car tires.

3.3.2 Wheels of requested rim width and flange contour and conforming to the dimensional requirements of a rim standardizing organization such as The Tire and Rim Association, Inc. The requested wheel is typically the highest volume usage rim width and flange for the test tire.

3.4 Test Time.

3.4.1 Car and Light Truck with Commercial Tires.

Calendar time:	60 days
Test hours:	1200 hours
Coordination hours:	4 hours

3.4.2 Compact Spare Tires.

Calendar time:	16 days
Test hours:	235 hours
Coordination hours:	4 hours

3.5 Test Required Information. An ATE Pre-test Specifications sheet as described in section 4.

3.6 Personnel/Skills.

3.6.1 Drivers need to be capable of safely operating a laden test vehicle under a variety of road conditions.

3.6.2 Drivers must be capable of visually inspecting tires for evidence of damage during the course of testing.

3.6.3 Drivers will need to be capable of accurately checking and maintaining proper inflation pressure.

3.6.4 Shop technicians must be capable of performing routine vehicle inspections, maintenance, ballasting, and wheel alignments.

3.6.5 Tire technicians must be capable of performing all measurements and data recording required within this procedure.

4 Procedure

4.1 Preparation.

4.1.1 ATE Pre-Test Specification Sheets.

4.1.1.1 Prepare and submit an ATE Pre-test Specification sheet to the GM test engineer and GM tire development engineer for all ATE tests. See Appendix A for a sample sheet.

4.1.1.2 The ATE Pre-test Specification sheet does not have to be approved prior to starting the test; however, if the test conditions differ significantly from those required by GM, the test may need to be restarted with new test pieces.

4.1.1.3 The ATE Pre-test Specification sheet will be reviewed for correct test conditions and approved by the GM test engineer and/or the GM tire development engineer.

4.1.1.4 The following information should be included on the ATE Pre-test Specification sheet.

4.1.1.4.1 Header. Tire supplier, GM test request number, supplier test number, and date.

4.1.1.4.2 Tire. Tire size, service description, construction number, load rating, serial numbers, and tire identification numbers.

4.1.1.4.3 Conditions. Load and inflation pressure for each test position on the vehicle.

4.1.1.4.4 Wheel. Wheel size and flange.

4.1.1.4.5 Vehicle. Vehicle fleet number, model description, model year, current mileage, engine, transmission, and alignment specifications for each axle/wheel position.

4.1.1.4.6 Timing. Estimated start and finish dates.

4.1.2 Tires and Wheels.

4.1.2.1 Install TPMS sensors on all test wheels.

4.1.2.2 Mount the test tires onto the wheels following a recommended practice such as that of the Rubber Manufacturer's Association.

4.1.2.3 Inflate the tire-wheel assembly to the test inflation pressure.

4.1.2.4 Balance the tire-wheel assemblies.

4.1.2.5 Measure tread depth in each groove in at least four equally spaced, circumferential locations around the tires. Average for each groove following ASTM F421.

4.1.2.6 Replacement tires should be prepared as necessary and stored until needed.

4.1.3 Vehicle.

4.1.3.1 Install four test tire-wheel assemblies on the test vehicle. Compact spares are tested on one position at a time with appropriately sized road tires on the other three positions.

4.1.3.2 Secure a spare tire, not a test tire and wheel assembly in the vehicle in the event a road hazard necessitates the temporary replacement of a test tire.

4.1.3.3 Position the driver's seat in the full rear position and place 68 kg of ballast on the driver's seat, simulating the weight of a typical driver.

4.1.3.4 Ballast the test vehicle to the specified test loads within ± 5 kg, securely anchoring all ballast. For compact spares, ballast the current test position to the specified test load while maintaining an equal load on both sides of the test axle.

4.1.3.5 Record all individual wheel loads.

4.1.3.6 Check front and rear wheel alignment with the vehicle ballasted and adjust to specifications if necessary. Record all alignment measurements.

4.1.3.6.1 The overall goal of setting wheel alignments for the purposes of this test is to minimize tire wear rates and thereby maximize the tires exposure to load cycles. Wheel alignments specifications for all test vehicles, therefore, are to optimize the alignment while driving down the road at 97 km/h.

4.1.3.6.2 The alignment specification will be nearly zero degrees toe and camber with the vehicle under test ballast conditions. This will result in the toe and camber being close to zero degrees while driving down the road at 97 km/h. Caster would be set to the manufacturer's specification as it has little influence on the tire wear rates.

4.1.3.6.3 The vehicle alignment, toe, and camber should be adjusted if necessary during the test to further minimize tire wear and maximize distance accumulation on the test vehicle.

4.1.3.6.4 Compact spare tests on the front axle will follow all vehicle alignment requirements except that the camber on the test position with the compact spare tire installed shall be $+1.5^\circ$.

4.1.3.7 Perform speedometer/odometer calibration with test tires on the vehicle and vehicle ballasted. Include this distance as the test distance on tires. A tachograph or a similar device is required for the

complete test to record the time, speed, and distance history.

4.1.3.8 Suspension and steering system components should be in good condition and properly adjusted at all times.

4.2 Conditions.

4.2.1 Environmental Conditions.

4.2.1.1 Ambient Temperature. Ambient temperature is known to influence the endurance properties of pneumatic tires; therefore, acceptable test locations should possess relatively warm ambient temperature with respect to the tires' intended usage market. The southwest Texas area is considered acceptable for tires used in the North American market.

4.2.1.2 Test Route.

4.2.1.2.1 The test route is designed such that a complete cycle or multiples thereof can be run within each shift, approximately 10 h.

4.2.1.2.2 A complete cycle will consist of the following road types, targeted speeds, and percentages of test distance as shown in Table 1.

Table 1: Test Cycle

Road Type	Target Speed (km/h)	Route Content (% Distance)
Paved Road with minimal cornering	105 to 115	58 to 60%
Paved Road with minimal cornering	90	36 to 38%
Gravel Road	40	3 to 4%

4.2.1.2.3 Test routes should be developed considering normal traffic volumes so that the traffic does not impede the test traffic.

4.2.1.2.4 The 90 km/h paved road should have some vertical content which will dynamically vary the loads on the tires.

4.2.1.2.5 The gravel content of the gravel road must be monitored closely. The quantity and type of aggregate used can impact the tire wear rates and potentially increase the risk of road hazards.

4.2.1.2.6 The gravel portion of each route cycle shall not be run when the gravel is wet as this will increase the potential for road hazards; however, if the gravel portion of a cycle is not run, it must be completed at a later time.

4.2.2 Test Conditions. Deviations from the requirements of the procedure shall have been agreed upon. Such requirements shall be specified on component drawings, test certificates, reports, etc.

4.2.2.1 Test Speeds. Test speeds are defined in Table 1 under the test route requirements.

4.2.2.2 Test Pressure. The test pressures refer to the test tire's cold inflation pressures.

Table 2: Test Pressures

Tire Type	Front Axle (kPa)	Rear Axle ^{Note 1} (kPa)
Passenger Car (T&RA)		
Standard Load and Light Load	180	240
Extra Load	220	280
Passenger Car (ETRTO)		
Standard Load	180	250
Reinforced	220	290
Compact Spare (T&RA)		
Standard Load and Light Load	420	420
Light Truck (T&RA)		
All Load Ranges (C, D, E) ^{Note 2}	250	Note 3

Note 1: Rear axle test pressure will be the minimum inflation pressure required to carry the maximum test load for the tire.

Note 2: Other load ranges may also be tested. If the specific load range is not listed in the table, the logic used to set the inflation pressure above should be extended to establish the necessary test pressure.

Note 3: Minimum inflation pressure to carry the test load for the tire. Typically this is the maximum pressure for the tire.

4.2.2.3 Test Load. The actual test loads are calculated based on the tire load capacities and the criteria in Table 3. Refer to the T&RA or ETRTO yearbooks for the specific load capacities of the test tires. Particular attention should be given to the standard referenced in Table 3, as the references are intentionally mixed within tire types.

4.2.2.3.1 Every effort shall be made to acquire a test vehicle capable of carrying the test loads in Table 3. When this cannot be accomplished, the following must be followed.

4.2.2.3.1.1 The test load and pressure of the rear axle shall be reduced in accordance with the applicable T&RA or ETRTO load formulas until the total load on the vehicle does not exceed the Gross Vehicle Weight (GVW) of the vehicle by more than 35 kg.

4.2.2.3.1.2 If while reducing the rear axle test load and pressure, the difference between the tire loads and the Gross Vehicle Weight (GVW) is 35 kg or less, the remaining difference can be deducted

from the front axle test load of the vehicle, without reducing the test inflation pressure of the front test tires.

Table 3: Test Loads

Tire Type	Front Axle ^{Note 1}	Rear Axle ^{Note 2}
Passenger Car (T&RA)		
Standard Load and Light Load	Maximum Load Capacity at 180 kPa	Maximum Load Capacity
Extra Load	Maximum Load Capacity at 220 kPa	Maximum Load Capacity
Passenger Car (ETRTO)^{Note 3}		
Standard Load	Maximum T&RA Load Capacity at 180 kPa	Maximum ETRTO Load Capacity
Reinforced	Maximum T&RA Load Capacity at 220 kPa	Maximum ETRTO Load Capacity
Compact Spare (T&RA)		
Standard Load and Light Load	85% of Maximum Load Capacity	95% of Maximum Load Capacity
Light Truck (T&RA)		
All Load Ranges (C, D, E)	Maximum Load Capacity at 250 kPa	Maximum Load Capacity

Note 1: Front axle loads are selected to be the maximum load capacity at or below the expected usage inflation pressures for the tires.

Note 2: Rear axle loads are selected to be at the rated load capacity for the tire.

Note 3: Through engineering judgement, the T&RA load formula was chosen to determine the intermediate load capacity at the test inflation pressure.

4.3 Instructions.

4.3.1 Tires are to be tested with all the same tire construction on all four positions on the vehicle at any given point in time and without tire rotations between wheel positions or changes in direction of rotation. There are two exceptions to this rule: 1) compact spares which will be described below, and 2) in cases where poor treadwear performance necessitates action prior to using a laboratory extension of this ATE procedure. If poor treadwear performance requires a rotation, GM will provide specific approval and the method must adhere to that described in 4.3.7 Laboratory Extension of ATE.

4.3.2 Compact spares are to be tested one tire at a time with at least one tire tested on the right front position and at least two tires tested on a rear position. The three non-test tires should have outside diameters representative of tires which could be reasonably expected to be used with the compact spare test tire on a production vehicle. As a general rule, the non-test tire diameter should be larger than the compact spare but not by more than 10%.

4.3.3 Tests are conducted in complete cycles, or multiples thereof, for each work shift. An allowable exception is the avoidance of the gravel section during rainy or other inclement weather. Maintain appropriate records of actual gravel road distance, and make up lost distance as required to stay within the specified tolerance for gravel road percentage found under the test route requirements. At no time should test mileage accumulation exceed 8000 km without accumulating the required gravel mileage. GM must approve any instances where this might not be feasible.

4.3.4 Wheel alignment must be checked with the ballasted vehicle at 8000 ± 1000 km intervals.

4.3.4.1 Correct the alignments as necessary.

4.3.4.2 Record all alignment measurements including those before and after adjustments.

4.3.5 Tire Inspection.

4.3.5.1 Visually inspect the test tires thoroughly at least once per day for any indication of structural degradation, road hazard, or pending tread wearout. More frequent inspections are required for compact spares and other tires nearing tread wearout.

4.3.5.2 All visual tire inspections, tread depth measurements, and photography are to be done with tires inflated to the test inflation pressure.

4.3.5.3 Monitor tire wear appearance during daily inspections. Correct as necessary any rapid wear with alignment adjustments.

4.3.5.4 Maintain the dynamic balance of all tires to minimize vehicle shake.

4.3.5.5 Measure tread depths in the same locations as done during the preparation at 8000 ± 1000 km intervals, following ASTM F421. The measurement interval for compact spares is 1600 ± 80 km.

4.3.5.6 Photographs are to be taken at 0 km, 24 000 km, 48 000 km, and end of test, 72 000 km, except compact spares that are photographed at every measurement interval. Photographs are to be taken of the most worn area of the tire, the area exhibiting structural degradation, or of any unusual

tire phenomena. Tire orientation in these photographs should appear as if looking at the tire from the rear of the vehicle.

4.3.5.7 Tire inflation pressure must be checked and recorded before and after each test cycle or work shift, and at least once more at approximately the middle of each shift. Care must be taken to not bleed air from the test tire during these measurements.

4.3.5.7.1 Never reduce the tire inflation pressure in a warm tire.

4.3.5.7.2 Only the cold inflation pressure can be adjusted to test pressure. See the definition of cold inflation pressure in the glossary.

4.3.5.7.3 If a warm tire is found below the test inflation pressure during or shortly after a test cycle, it should be removed from the test vehicle and replaced with a non-test spare until the cause for the low inflation pressure is determined.

4.3.5.7.4 If a tire is repeatedly found low, or if a low inflation pressure reading cannot be explained by a lower ambient temperature of approximately 6 kPa per 5°C, then remove the tire until the root cause is determined.

4.3.6 Test Tire Removal.

4.3.6.1 Test tires except compact spares will continue test distance accumulation until at least four (4) tires are removed from the test. Tires will be removed from test for one of the following reasons, whichever occurs first for each tire:

- Accumulation of 72 000 km of test distance
- Tread wearout
- Structural degradation

4.3.6.2 Compact spare tires must continue test distance accumulation until at least three (3) tires are removed from the test. Compact spare tires will be removed from test for one of the following reasons, whichever occurs first for each tire:

- Accumulation of 4850 km of test distance
- Tread is worn smooth across at least one groove
- Cord material is visible
- Structural degradation

4.3.6.3 A tire removed because of a non-repairable road hazard may not be allowed in the data analysis and, therefore, it may not be counted as one of the required samples. See 5.1.2 Weibull Calculations for additional information.

4.3.6.4 Tires are removed from test individually as soon as a removal condition noted above or a non-repairable road hazard occurs.

4.3.6.5 Precautions must be made to remove tires before gross air loss occurs and to prevent running tires in an under-inflated condition. This is to insure that the tires can be inspected after the test is complete. This post-test inspection of the tires is an important element of this test; therefore, additional tires may need to be tested if an insufficient number of tires exist for inspection.

4.3.6.6 Every attempt, allowed by this procedure, should be made to complete ATE testing outdoors until the tire accumulates the required test distance or until structural degradation dictates removal from the test. However, specific tire designs may not possess the treadwear performance to facilitate the accumulation of the required test distance. In these cases, the tire is to be removed in a state of treadwear to facilitate the removed tire continuing test distance accumulation in a laboratory environment as detailed below.

4.3.6.7 If a tire is removed for any reason during a test cycle, the test driver must record all pertinent information at the time of removal from the vehicle:

- a. Inflation pressure, immediately upon stopping the vehicle
- b. Rate of air loss
- c. Speed when tire condition was noticed
- d. Vehicle odometer reading
- e. Cause of condition requiring removal, if known
- f. Complete tire identification
- g. Location in the test cycle
- h. Any other unusual or noteworthy comments
- i. Photographs, which may be taken at a later time
- j. Cold inflation pressure, once the tire has been allowed to cool

4.3.6.8 Road Hazards.

4.3.6.8.1 A tire with a puncture type road hazard may be repaired and then returned to test distance accumulation provided the following conditions are all satisfied.

- a. The tire has accrued 8000 km or more test distance prior to the road hazard.
- b. The puncture is repairable as defined by the Rubber Manufacturer's Association.

4.3.6.8.2 The recommended puncture repair technique is to plug the puncture hole and apply a patch to the inside of the tire.

4.3.6.8.3 If a road hazard is repaired, maintain complete documentation and reinstall the tire on the vehicle to continue the test distance accumulation. Documentation shall include the test

distance at occurrence, lost test distance, type of road hazard, photos, etc.

4.3.6.8.4 If a tire receiving a road hazard cannot be repaired, it must be removed from test and a replacement test tire must be installed on the vehicle.

4.3.6.9 Permanent Tire Removal.

4.3.6.9.1 If a tire cannot continue test distance accumulation, either on the outdoor test or on a laboratory extension of the ATE test as described in 4.3.7 Laboratory Extension of ATE, then it will be considered a permanent removal.

4.3.6.9.2 Perform a complete inspection of the permanently removed tire as described in 4.3.5 Tire Inspection. This inspection must include tread depths and photographs showing the tread profile. Additional photographs are required to record the structural degradation or other phenomenon causing the tire to be permanently removed.

4.3.7 Laboratory Extension of ATE.

4.3.7.1 A tire removed from testing due to pending tread wearout may be required to continue mileage accumulation on a GM approved indoor laboratory ATE supplement until the end of test mileage, 72 000 km. This is solely at the discretion of GM.

4.3.7.2 Every attempt allowed by this procedure should be made to complete ATE testing outdoors.

4.3.7.2.1 Tires are not to be rotated between positions on the vehicle or flipped on the wheel to increase the test distance accumulation without the specific approval of the GM test engineer and the GM tire development engineer.

4.3.7.2.2 If the approval above is granted, the only tire rotation method permissible to increase the outdoor test distance accumulation is to flip the tire on the wheel and move it to the opposite side of the vehicle on the same axle.

4.3.7.3 After all attempts to accumulate test distance outdoors is exhausted, the tire should be moved into the laboratory; however, the tire must also meet the following criteria before being considered for a continued test distance accumulation on a GM approved laboratory extension of the ATE test.

4.3.7.3.1 The tire must be removed from outdoor testing before any reinforcement material is visible. There must be enough tread remaining to facilitate indoor testing.

4.3.7.3.2 All tires removed for treadwear require GM approval before continuing on an indoor laboratory ATE supplement.

5 Data

5.1 Calculations.

5.1.1 Wear Life Calculations.

5.1.1.1 Wear life may be determined by either of the two tread wearout conditions described in the glossary. The wear life resulting from those two conditions are calculated differently.

5.1.1.2 The wear life of a tire is to be determined by the condition which yields the minimum distance to tread wearout.

5.1.1.3 In the shoulder regions, the wear life will be equal to the accumulated test distance when the carcass material is exposed.

5.1.1.4 In the interior portion of the tire, the tire will be at tread wearout when the tread is worn to the wear indicators as defined in Tread Wearout. The method used to determine the wear life will depend on the wear state.

5.1.1.4.1 If the wear state of the tire meets the tread wearout definition in the glossary, then the wear life will be calculated using a straight-line interpolation between the last two measurement intervals. The last two average groove depths for each groove and the associated test distances will be used to interpolate to the wear indicator height of 1.59 mm.

$$TL = D_1 + (GD_1 - h) * [(D_2 - D_1) / (GD_2 - GD_1)]$$

Where:

TL = Interpolated Wear Life, in km

D₁ = First Test Distance, in km

D₂ = Second Test Distance, in km

GD₁ = Average Groove Depth at first distance, in mm

GD₂ = Average Groove Depth at second distance, in mm

h = Height of Wear Indicator, 1.59 mm

5.1.1.4.2 If the wear state of the tire does not meet the definition of tread wearout, then calculate the projected tread wear life for each groove, similar to ASTM F1016.

5.1.1.4.2.1 Projected wear life is to be calculated assuming a linear wear rate without considering a break-in period for the test tires. Determine the linear wear rate by regressing the dependent variable, average groove depth, and the independent variable, test distance. The linear wear rate (WR) will be the slope of this regression. The resulting line will be:

$$GD_{(Average)} = GD_{(Intercept)} - WR(Test Distance)$$

Where:

GD_(Average) = average Groove Depth, in mm

GD_(Intercept) = y-intercept from regression, in mm

WR = regression slope, Wear Rate, in mm/km

5.1.1.4.2.2 The projected wear life for each groove can now be estimated using the slope and intercept calculated above.

$$TL = (1.59 \text{ mm} - GD_{(\text{Intercept})}) / WR$$

Where:

TL = Projected Wear Life, in km

GD_(Intercept) = y-intercept from regression, in mm

WR = regression slope, Wear Rate, in mm/km

5.1.1.4.2.3 The wear life of the tire is now defined as the minimum of the projected wear lives calculated above.

5.1.1.4.2.4 The average wear life of the tire is the average of projected wear lives calculated above.

5.1.2 Weibull Calculations (Except Compact Spares).

5.1.2.1 An analysis based on Weibull distribution theory is used to quantify all test results except those results for compact spare tires.

5.1.2.2 If all four tires reach the end-of-test mileage without structural degradation or tread wearout, then no Weibull Analysis is to be performed.

5.1.2.3 All tires with a final removal cause classified as a structural degradation or tread wearout shall be considered expired items for the Weibull analysis.

5.1.2.3.1 A repaired tire with subsequent structural degradation may be classified either as a road hazard or structural degradation, depending on the proximity of road hazard location, type of degradation, and dissection analysis of the cause of removal.

5.1.2.4 If a minimum of three expired items are not obtained from the preceding step, end-of-test tires and road hazard tires with test distances greater than 50 000 km shall be considered expired items.

5.1.2.5 Any tire not classified as expired shall be classified as a suspended item for the analysis. Suspended items, end-of-test tires, and other non-failure removals with test distances lower than the first expired item will not be used in the Weibull analysis unless required to obtain the minimum of three items needed for the analysis.

5.1.2.6 No distinction shall be made between test positions, although an equal number of front and rear position tires are desirable.

5.1.2.7 Perform the Weibull analysis, determining the B10 Life and B50 Life for the test tires.

5.1.2.7.1 The B10 Life and B50 Life are to be calculated using only the outdoor test distances.

5.1.2.7.2 If applicable, the B10 Life and B50 Life are then to be calculated for the combined test

distances accumulated both outdoors and in the laboratory.

5.2 Interpretation of Results. Not applicable.

5.3 Test Documentation.

5.3.1 Reporting Interval. Tire test reports are to be provided to GM at 24 000 km, 48 000 km, end-of-test, and after any tire removal.

5.3.2 For TIP approved data being uploaded to the GM Global Tire Database, provide data at each Reporting Interval following the electronic file formats given in Appendix B. The image file will contain a report as shown in Appendix D. Otherwise, document the test results as described below.

5.3.3 Provide the following reports for data not being uploaded to the GM Global Tire Database at each Reporting Interval.

5.3.3.1 A summary report as shown in Appendix C.

5.3.3.2 Detailed test reports including the following minimum information described below, following the report template as shown in Appendix D.

5.3.3.2.1 The interim reports are to contain at least the following information for each test tire.

- a. Test position on vehicle
- b. Test load
- c. Temperature
- d. Test tire inflation pressure
- e. Test rim width
- f. Total test distance
- g. Reason for removal and complete description of tire condition
- h. Details of any road hazard repairs, including test distance at occurrence and location and type of road hazard.
- i. Cold inflation pressure after removal.
- j. Individual groove depth measurements for each measurement interval.
- k. Projected distance at tread wearout
- l. Photographs

5.3.3.2 The end-of-test reports will include all the information for the interim reports and this additional information for each test tire.

- a. B10 Life and B50 Life for outdoor only test distance
- b. B10 Life and B50 Life for combined outdoor and laboratory test distances
- c. Average wear life for a compact spare test
- d. Summary of the visual inspection at end-of-test
- e. Conclusions
- f. General comments

- g. Temperature report
- h. Tread wearout projections
- i. Individual wheel loads at each measurement
- j. Front axle load at each measurement
- k. Rear axle load at each measurement
- l. Groove depths tabulation
- m. Groove depth charts
- n. Tread wearout progression charts
- o. Tread wearout projection charts
- p. Final photographs

6 Safety

This standard may involve hazardous materials, operations, and equipment. This standard does not propose to address all the safety problems associated with its use. It is the responsibility of the user of the standard to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.

7 Notes

7.1 Glossary.

Average Wear Life: When average wear lives are reported, the average for each axle, i.e., front and rear will be computed first, and the overall average will be the average of the individual axle averages.

Carcass Material: Any material manufactured into the tire that is not intended as a wearable surface such as body cords, steel belts, nylon overlays, etc.

Cold Inflation Pressure: The pressure of the contained air mass within a tire that has been allowed to thermally stabilize to the prevailing ambient temperature. A tire may be considered stabilized after remaining at rest for a minimum of 1.5 h while shielded from direct sunlight.

End of Test Tire: Tire removed at test completion that has not been classified for Weibull analysis as an expired item, tread wearout, or road hazard removal.

Maximum Load: The load capacity at the maximum required test pressure for the tire being tested. Refer to the most recent *T&RA* or *ETRTO Yearbook* or other standard for specific values, as applicable.

Minimum Load: The load capacity at the minimum required test pressure for the tire being tested. Refer to the most recent Tire & Rim Association or *ETRTO Yearbook* for specific values, as applicable.

Road Hazard: An event or tire condition typically including, but not limited to: punctures, cuts, and impact breaks.

Structural Degradation: Loss of integrity of the tire structure as evidenced by, but not limited to loss of adhesion, blisters or separations, tread or other material loss, loss of air, and cracks to cord material.

Tread Wearout: A tire condition exhibiting either of the following:

- Exposure of cord or belt material due to the wearing away of rubber surface material.
- Tread material worn even with the treadwear indicators in at least three places in each of two adjacent grooves

Wear Life: The accumulated test distance to tread wearout.

Weibull Analysis:

- B10 Life: 10th percentile for a Weibull distribution calculated using 2-parameter, median ranks with distance as the independent variable.
- B50 Life: 50th percentile for a Weibull distribution calculated using 2-parameter, median ranks with distance as the independent variable.

7.2 Acronyms, Abbreviations, and Symbols.

ASTM	American Society (for) Testing (and) Materials
ATE	Accelerated Tire Endurance
CTS	Component Technical Specifications
ETRTO	European Tire (and) Rim Technical Organization
GVW	Gross Vehicle Weight
T&RA	Tire & Rim Association
TPMS	Tire Pressure Monitoring System

8 Coding System

This standard shall be referenced in other documents, drawings, VTS, CTS, etc. as follows:

Test to GMW15001

9 Release and Revisions

9.1 Release. This standard originated in January 2006, replacing T97ATE1. It was first approved by the Tire and Wheel Global Subsystem Leadership Team in August 2006. It was first published in February 2007.

Appendix A - ATE Pre-Test Specification (Sample)

GENERAL MOTORS ATE PRE-TEST SPECIFICATIONS

Date Prepared
TWS Tire Work Request #
Test Purpose
Tire Supplier
Test Site

Test Number (Optional)
Project Number (Optional)

TIRE INFORMATION

Tire Size	Rim Information		Front		Rear	
	Descrip	Material	Inflation	Load	Inflation	Load
Position	DOT		Tire ID (Serial #)		Construction #	
Left Front						
Right Front						
Left Rear						
Right Rear						
Spare						
Spare						

TEST INFORMATION

Test Route ID:

Test Vendor

Project Engineer
Phone Number
Test Coordinator
Phone Number

Tire Supplier

Project Engineer
Phone Number
Test Coordinator
Phone Number

Timing

Est Start Date
Est Finish Date

TEST VEHICLE

Vehicle ID

Make
Model
Model Year
Mileage
Transmission
Engine

Suspension Alignment (Degrees)

Suspension Type	Position	Toe	Camber	Caster
	Front			
	Left			
	Right			
	Rear			
	Left			
	Right			

Appendix B - Electronic File Format Definitions for GM Global Tire Database Submission

.TPC and .DAT File Naming Convention – Accelerated Tire Endurance				
File Type	File Name	Explanation	Example	
TPC	XXFMDDSS.TPC	Z – Force & Moment Test M – Month (1-9 and A,B,C) DD – Day of Month SS – Test Number (ex. 01)	XXZA0426.TPC ATE Test (Third Character = Z) Tested on November 4 ("B" = November, "04" = Test Day) Test Number 26	
Line #	Variable	Char. Limit	Example	Comment
GENERAL INFORMATION				
1	TPC File Name	12C	xxZxxxxx.TPC	Name of this *.TPC file
2	Data File Name	12C	xxZxxxxx.DAT	Associated data file name
3	Image File Name	12C	xxZxxxxx.PDF	Image file name of ATE Report
4	Tire Test Request Number	8C	90-12345	Supplied by GM
5	Size	13C	LT225/75R16	
6	Load Rating	2C	SL	Valid Codes Only (see GM Global Tire Database
7	Supplier	3C	XXX	Valid Codes Only (see GM Global Tire Database
8	Construction Number	12C	1234567890AB	
9	Rim Width	5.2	10.50	(inches)
10	Front Inflation	3.0	123	(kPa)
11	Front Load	5.0	12345	(Newton)
12	Rear Inflation	3.0	123	(kPa)
13	Rear Load	5.0	12345	(Newton)
14	Test Site	3C	XXX	Valid Codes Only (see GM Global Tire Database
15	Tester	3C	XXX	Valid Codes Only (see GM Global Tire Database
16	Test Date	10C	01/20/1991	(Month/Day/Year)
17	Comments	30C	Development	
TEST RESULTS				
18	Software Title	32C	Weibull Analysis, Inc	
19	Software Version	16C	V8.02.35	
20	Sample Size	F1.0	4	(Number of tires in analysis)
21	B10 Life	F6.0	123456	(kilometer)
22	B50 Life	F6.0	123456	(kilometer)
23	Characteristic Life	F6.0	123456	(kilometer)
24	Slope	F5.2	12.34	
25	Correlation	F6.4	1.2345	
TEST TIRE DATA				
(Note: The following structure to be repeated for each tire tested. Reference Line 20 for number of tires)				
26	Tire ID	10C	12345ABCDE	
27	DOT Serial	6C	AB1234	
28	Position	2C	AB	(LF, RF, LR, or RR)
29	Distance	6.0	123456	
30	Removal Code	3C	ABC	(Valid Codes Only)
31	Percent Worn, outside	F3.0	123	
32	Percent Worn, inside	F3.0	123	
33	Treadwear Description	128C		Free Flow Text Field, 128 Characters in Length
34	Bead	128C		Free Flow Text Field, 128 Characters in Length
35	Sidewall	128C		Free Flow Text Field, 128 Characters in Length
36	Liner	128C		Free Flow Text Field, 128 Characters in Length
37	Road Hazard	128C		Free Flow Text Field, 128 Characters in Length
38	Comment	256C		Free Flow Text Field, 256 Characters in Length

Appendix C - Sample GM Global Database Summary Report - Page 1/2

ACCELERATED TIRE ENDURANCE

GM PROPRIETARY

Size/LR/Serv Descr: P245/70R16 / SL / 112S

Supplier: BFG

Construction: KING-DATA

Tire Type Code: PARPS

Rim Width: 7

Front Inflation [kPa]: 123

Front Load [N]: 12345

Work Request: 01- 345

Requestor: ELN

Program: GMT360

Test Site/Tester: MPG / TWS

Test Date: 07/27/2001

Status: A

Rear Inflation [kPa]: 345

Rear Load [N]: 67890

Purpose/Comment: FOR DATABASE ONLY - NO REAL TE

Weibull Analysis

Sample Size	B10 Life [km]	B50 Life [km]	Characteristic Life [km]	Slope	Correlation
4	123456	654321	454545	12.34	0.1234

Software Title XXX/Version: WEIBUL ANALYSIS BY ABC, INC.123 / VERSION V8.02.3

Shoulder, Percent Worn

PDF	Tire ID	Position	DOT	Removal Code	Distance [km]	Inside	Outside
PDF	TIRE3.BCDE	LF	AB1234	A02	123456	195	185
PDF	TIRE2.BCDE	LF	AB1234	A02	123456	195	185
PDF	TIRE1.BCDE	LF	AB1234	A02	123456	195	185
PDF	TIRE4.BCDE	LF	AB1234	A02	123456	195	185

Physical Tire Inspection Comments

Tire ID:

TIRE3.BCDE

Work Request: 01- 345

Treadwear	TREADWEAR DISCRIPTION, I HAVE 128 CHARACTERS TO USE HERE AND THIS SHOULD BE MORE THAN ENOUGH ROOM FOR DISCRIPTIE WORDS.ADADFFD
Bead	BEAD AREA DISCRIPTION, I HAVE 128 CHARACTERS TO USE HERE AND THIS SHOULD BE MORE THAN ENOUGH ROOM FOR DISCRIPTIE WORDS.ADAFSDF
Sidewall	SIDEWALL. DISCRIPTION, I HAVE 128 CHARACTERS TO USE HERE AND THIS SHOULD BE MORE THAN ENOUGH ROOM FOR DISCRIPTIE WORDS.ADFADFF
Liner	LINER.... DISCRIPTION, I HAVE 128 CHARACTERS TO USE HERE AND THIS SHOULD BE MORE THAN ENOUGH ROOM FOR DISCRIPTIE WORDS.ASDFADF
Road Hazard	ROAD HAZARDISCRIPTION, I HAVE 128 CHARACTERS TO USE HERE AND THIS SHOULD BE MORE THAN ENOUGH ROOM FOR DISCRIPTIE WORDS.ADADFAD
Comment	COMMENT....., I HAVE 128 CHARACTERS TO USE HERE AND THIS SHOULD BE MORE THAN ENOUGH ROOM FOR DISCRIPTIE WORDS.COMMENT.. CONTINUING , I HAVE 128 CHARACTERS TO USE HERE AND THIS SHOULD BE MORE

Appendix C Sample GM Global Database Summary Report - Page 2/2

THAN ENOUGH ROOM FOR DISCRIPTIE WORDS.ADADASDADSFAAAAAA

Tire ID:**TIRE2.BCDE****Work Request:** 01- 345

Treadwear	TREADWEAR DISCRIPTION, I HAVE 128 CHARACTERS TO USE HERE AND THIS SHOULD BE MORE THAN ENOUGH ROOM FOR DISCRIPTIE WORDS.ADADFFD
Bead	BEAD AREA DISCRIPTION, I HAVE 128 CHARACTERS TO USE HERE AND THIS SHOULD BE MORE THAN ENOUGH ROOM FOR DISCRIPTIE WORDS.ADAFSDF
Sidewall	SIDEWALL. DISCRIPTION, I HAVE 128 CHARACTERS TO USE HERE AND THIS SHOULD BE MORE THAN ENOUGH ROOM FOR DISCRIPTIE WORDS.ADFADFF
Liner	LINER.... DISCRIPTION, I HAVE 128 CHARACTERS TO USE HERE AND THIS SHOULD BE MORE THAN ENOUGH ROOM FOR DISCRIPTIE WORDS.ASDFADF
Road Hazard	ROAD HAZARDISCRIPTION, I HAVE 128 CHARACTERS TO USE HERE AND THIS SHOULD BE MORE THAN ENOUGH ROOM FOR DISCRIPTIE WORDS.ADADFAD
Comment	COMMENT....., I HAVE 128 CHARACTERS TO USE HERE AND THIS SHOULD BE MORE THAN ENOUGH ROOM FOR DISCRIPTIE WORDS.COMMENT.. CONTINUING , I HAVE 128 CHARACTERS TO USE HERE AND THIS SHOULD BE MORE THAN ENOUGH ROOM FOR DISCRIPTIE WORDS.ADADASDADSFAAAAAA

Tire ID:**TIRE1.BCDE****Work Request:** 01- 345

Treadwear	TREADWEAR DISCRIPTION, I HAVE 128 CHARACTERS TO USE HERE AND THIS SHOULD BE MORE THAN ENOUGH ROOM FOR DISCRIPTIE WORDS.ADADFFD
Bead	BEAD AREA DISCRIPTION, I HAVE 128 CHARACTERS TO USE HERE AND THIS SHOULD BE MORE THAN ENOUGH ROOM FOR DISCRIPTIE WORDS.ADAFSDF
Sidewall	SIDEWALL. DISCRIPTION, I HAVE 128 CHARACTERS TO USE HERE AND THIS SHOULD BE MORE THAN ENOUGH ROOM FOR DISCRIPTIE WORDS.ADFADFF
Liner	LINER.... DISCRIPTION, I HAVE 128 CHARACTERS TO USE HERE AND THIS SHOULD BE MORE THAN ENOUGH ROOM FOR DISCRIPTIE WORDS.ASDFADF
Road Hazard	ROAD HAZARDISCRIPTION, I HAVE 128 CHARACTERS TO USE HERE AND THIS SHOULD BE MORE THAN ENOUGH ROOM FOR DISCRIPTIE WORDS.ADADFAD
Comment	COMMENT....., I HAVE 128 CHARACTERS TO USE HERE AND THIS SHOULD BE MORE THAN ENOUGH ROOM FOR DISCRIPTIE WORDS.COMMENT.. CONTINUING , I HAVE 128 CHARACTERS TO USE HERE AND THIS SHOULD BE MORE THAN ENOUGH ROOM FOR DISCRIPTIE WORDS.ADADASDADSFAAAAAA

Tire ID:**TIRE4.BCDE****Work Request:** 01- 345

Treadwear	TREADWEAR DISCRIPTION, I HAVE 128 CHARACTERS TO USE HERE AND THIS SHOULD BE MORE THAN ENOUGH ROOM FOR DISCRIPTIE WORDS.ADADFFD
Bead	BEAD AREA DISCRIPTION, I HAVE 128 CHARACTERS TO USE HERE AND THIS SHOULD BE MORE THAN ENOUGH ROOM FOR DISCRIPTIE WORDS.ADAFSDF
Sidewall	SIDEWALL. DISCRIPTION, I HAVE 128 CHARACTERS TO USE HERE AND THIS SHOULD BE MORE THAN ENOUGH ROOM FOR DISCRIPTIE WORDS.ADFADFF
Liner	LINER.... DISCRIPTION, I HAVE 128 CHARACTERS TO USE HERE AND THIS SHOULD BE MORE THAN ENOUGH ROOM FOR DISCRIPTIE WORDS.ASDFADF
Road Hazard	ROAD HAZARDISCRIPTION, I HAVE 128 CHARACTERS TO USE HERE AND THIS SHOULD BE MORE THAN ENOUGH ROOM FOR DISCRIPTIE WORDS.ADADFAD
Comment	COMMENT....., I HAVE 128 CHARACTERS TO USE HERE AND THIS SHOULD BE MORE THAN ENOUGH ROOM FOR DISCRIPTIE WORDS.COMMENT.. CONTINUING , I HAVE 128 CHARACTERS TO USE HERE AND THIS SHOULD BE MORE THAN ENOUGH ROOM FOR DISCRIPTIE WORDS.ADADASDADSFAAAAAA

Appendix D - Sample Detailed Test Report Template - Page 1/26

COMPANY NAME*Division***COMPANY NAME***Location***G.M. 'ATE' DURABILITY COMPLETED TEST REPORT****G.M. 'ATE' DURABILITY FINAL INSPECTION REPORT****PROGRAM:** Size / Tire Name, CONSTR. NO. XXX**GM Work Request #: XX-XXX**
(REG. #: XXXXX)**ENGINEER:** Name**DATE:** Date report is compiled

The ATE test of the subject tire, test number XXXX, completed testing at XXXX (location) on XXXXX date.

The attached data packet contains the completed test report for this ATE test.

Contents:

- Test Summary Page
 - Weibull Analysis
 - Tire Test Summary
 - Total test distance
 - Removal reason
 - Projected km to wear out
 - Rim
 - Inflation
 - Cold inflation at removal
 - Average test load
 - Temperature
- Visual Inspection at Test Completion
- Conclusions
- General Comments
- Temperature Report
- Load History Report
- Wear Out Projections Report
- Final Vehicle Alignment Report
- Skid Depth Report
- Groove Depth Charts
- Wear Out Progression Charts
- Wear Out Projection Charts
- Final Photographs

Appendix D - Sample Detailed Test Report Template - Page 2/26

COMPANY NAME

COMPANY NAME

G.M. 'ATE' DURABILITY COMPLETED TEST REPORT

Division

Location

Project Registration Number

XXXX

Test Number

XXXXXX

G.M. Program	200X MY GMXXXX
Tire Size / Service Description	XXXX
Tire Name	XXXX
Tire Type	XXX
O.E. Construction Number	XXXXX

WEIBULL ANALYSIS			Classification
Category	Requirement	Test Results	Judgement
B10	XXXX km	XXXX km	XXX
B50	XXXX km	XXXX km	XXX

TIRE TEST SUMMARY									
TIRE ID	TEST POS.	TOTAL TEST DISTANCE (KM)	REMOVAL REASON	PROJECTED DISTANCE TO WEAROUT (KM)	RIM WIDTH (IN.)	AVG TEST LOAD (N)	TEST INFL. (KPA)	COLD INFL @ COMP. (KPA)	AMBIENT TEMP., °C
DDD	RR	72.420	X						
CCC	LR	72.420	X						
BBB	RF	72.420	X						
AAA	LF	72.420	X						
Note: All mileage on vehicle unless noted otherwise.									

Conclusions:

- XXX
- XXX

Appendix D – Sample Detailed Test Report Template – Page 3/26

COMPANY NAME *Division*
COMPANY NAME *Location*
G.M. 'ATE' DURABILITY COMPLETED TEST REPORT

Project Registration Number XXXX Test Number XXXXXX

TIRE INSPECTION					INSPECTION RESULTS		
TIRE ID	POS.	TOTAL TEST DIST. (KM)	REMOVAL REASON	INTERIOR	SIDEWALLS	TREAD/BELTS	
DDD	RR	XXX	XXX	• XX •	• XX •	•	XX
CCC	LR	49,299	XXX	• XX •	• XX •	•	XX
BBB	RF	72,420	XXX	• XX •	• XX •	•	XX
AAA	LF	72,420	XXX	• XX •	• XX •	•	XX
Note: XXX							

Summary of Findings:

- XXXX

Appendix D- Sample Detailed Test Report Template - Page 4/26

COMPANY NAME

COMPANY NAME

G.M. 'ATE' DURABILITY COMPLETED TEST REPORT

XXX Division

XXX Location

General Comments Report

Project Registration

Test Number

Spec Code

XXX

XXX

XX

COMMENT DATE

MEAS SEQ

GENERAL COMMENT

Year-month-day

XXXX-XX-XX

XX

comments

Appendix D - Sample Detailed Test Report Template - Page 5/26

COMPANY NAME		XXX Division		Spec Code		XX	
COMPANY NAME		XXX Location		XX			
G.M. 'ATE' DURABILITY COMPLETED TEST REPORT		Temperature Report					
Project Registration		Test Number					
XX		XX					
MP	MP	AVG	AVG	TEST	MIN	MAX	
START	END	TEMP	TEMP	LOCATION	MEASURE	MEASURE	
DATE	DATE	DEG C	DEG C	DATE	DATE	DATE	
Yr-mnth-day	Yr-mnth-day	XX	XX	X	Yr-mnth-day	Yr-mnth-day	
XXXX-XX-XX	XXXX-XX-XX	XX	XX		XXXX-XX-XXXXXX-XX-XX	XXXX-XX-XXXXXX-XX-XX	
MIN		MAX		AVG		MIN	
Yr-mnth-day		Yr-mnth-day		X		X	

Appendix D - Sample Detailed Test Report Template - Page 6/26

COMPANY NAME

COMPANY NAME

G.M. 'ATE' DURABILITY COMPLETED TEST REPORT

XXX Division

XXX Location

Load History Report (Newtons)

Test Number

Spec Number

Spec Code

Vehicle

Project Registration

Tire Size

XX

XX

XX

XX

XX

XX

TEST

KILOMETERS

0

8047

16093

24140

32187

40234

48280

56327

64374

72420

LF

XXXX

RF

XXXX

LR

XXXX

RR

XXXX

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Appendix D - Sample Detailed Test Report Template - Page 7/26

COMPANY NAME		XXX Division	
COMPANY NAME		XXX Location	
G.M. 'ATE' DURABILITY COMPLETED TEST REPORT		Wearout Projections Report	
Project Registration	XX	Test Number	XX
Tire Size	XX	Spec Number	XX
		Spec Code	XX

TIRE NO	XX	AXLE	TEST KILOMETER	PROJECTED WEAROUT (KM)
AAA	L	F	72420	X
BBB	R	F	72420	↓
CCC	L	R	72420	
DDD	R	R	72420	

Appendix D - Sample Detailed Test Report Template - Page 8/26

COMPANY NAME

COMPANY NAME

G.M. 'ATE' DURABILITY COMPLETED TEST REPORT

XXX Division

XXX Location

Alignment Report

Test Number

Spec Number

Spec Code

XX

Project Registration

XXX

Tire_Size

XXX

Vehicle

XXX

TIRE
NO
AAA

INSPECTION
XX
72420

VEHICLE
POSITION
LFH

CAMBER
DEG
XXX

CASTOR
DEG
XXX

TOE
DEG
XX

TYPE
ALIGN
XX

MEAS
DATE
Yr-mnth-day

BBB

72420

RFH

XXX

XXX

XX

XX

Yr-mnth-day

CCC

72420

LRH

XXX

XXX

XX

XX

Yr-mnth-day

DDD

72420

RRH

XXX

XXX

XX

XX

Yr-mnth-day

Appendix D - Sample Detailed Test Report Template - Page 9/26

COMPANY NAME		XXX Division		Spec Code		XX	
COMPANY NAME		XXX Location					
G.M. 'ATE' DURABILITY COMPLETED TEST REPORT		Skid Report					
Project Registration		XX		Test Number		XX	
Test Code		XX		Test Name		XX	
Tire Size		XX		Spec Number		XX	

TIRE NO	INSPECTION XX	VEHICLE POSITION	ISS SKID DEPTH MM	ISH SKID DEPTH MM	CTR SKID DEPTH MM	OSH SKID DEPTH MM	OSS SKID DEPTH MM	?? SKID DEPTH MM
DDD	0	LH						
DDD	8047	LH						
DDD	16093	LH						
DDD	24140	LH						
DDD	32187	LH						
DDD	40234	LH						
DDD	48280	LH						
DDD	56327	LH						
DDD	64374	LH						
DDD	72420	LH						
CCC	0	RF						
CCC	8047	RF						
CCC	16093	RF						
CCC	24140	RF						
CCC	32187	RF						
CCC	40234	RF						
CCC	48280	RF						
CCC	64374	RF						
CCC	72420	RF						
BBB	0	LR						
BBB	8047	LR						
BBB	16093	LR						
BBB	24140	LR						
BBB	32187	LR						
BBB	40234	LR						
BBB	48280	LR						
BBB	56327	LR						
BBB	64374	LR						
BBB	72420	LR						

skid depth measurement here

skid depth measurement here

skid depth measurement here

Appendix D - Sample Detailed Test Report Template - Page 10/26

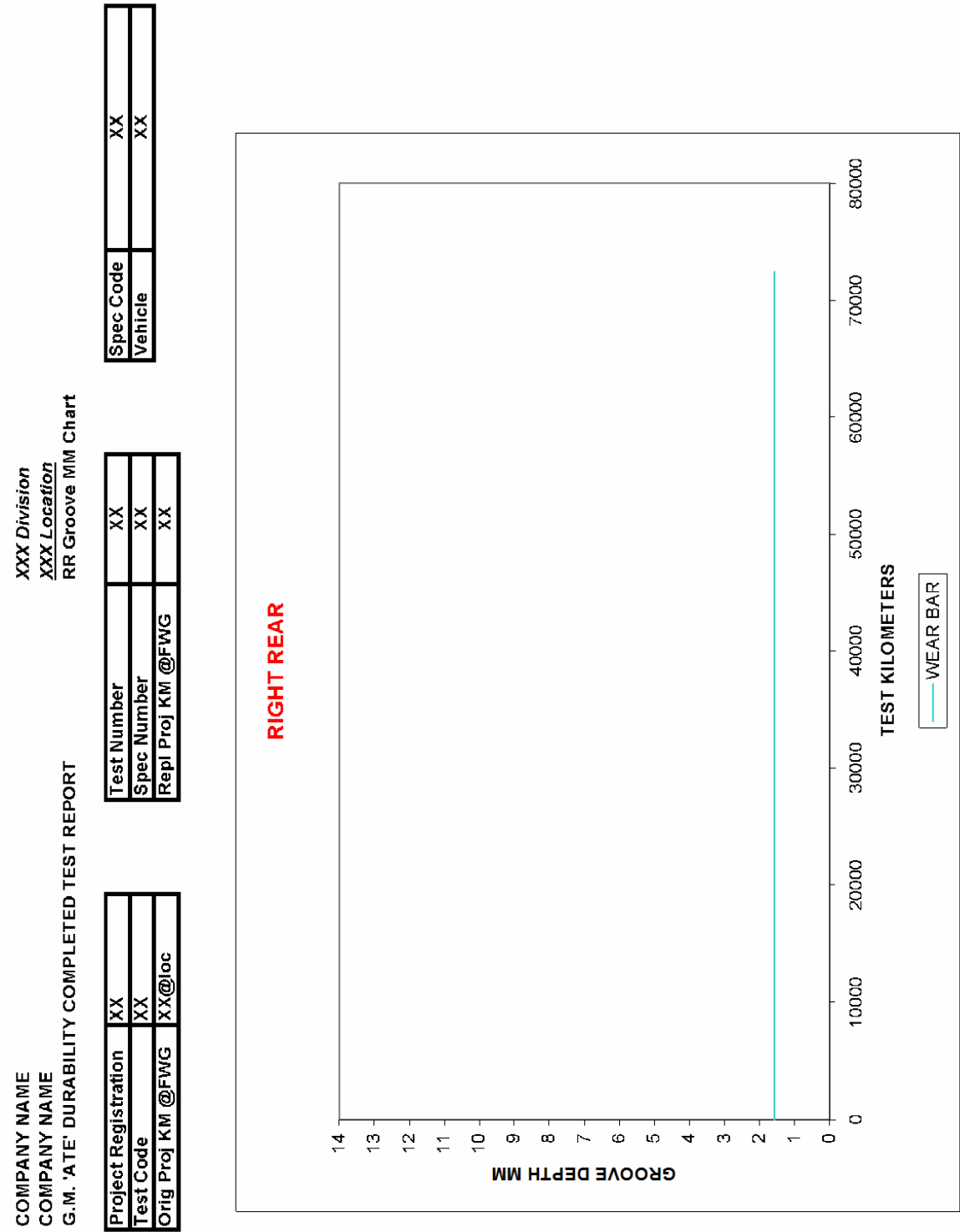
skid depth measurement here

RRH
RRH
RRH
RRH
RRH
RRH
RRH
RRH
RRH
RRH

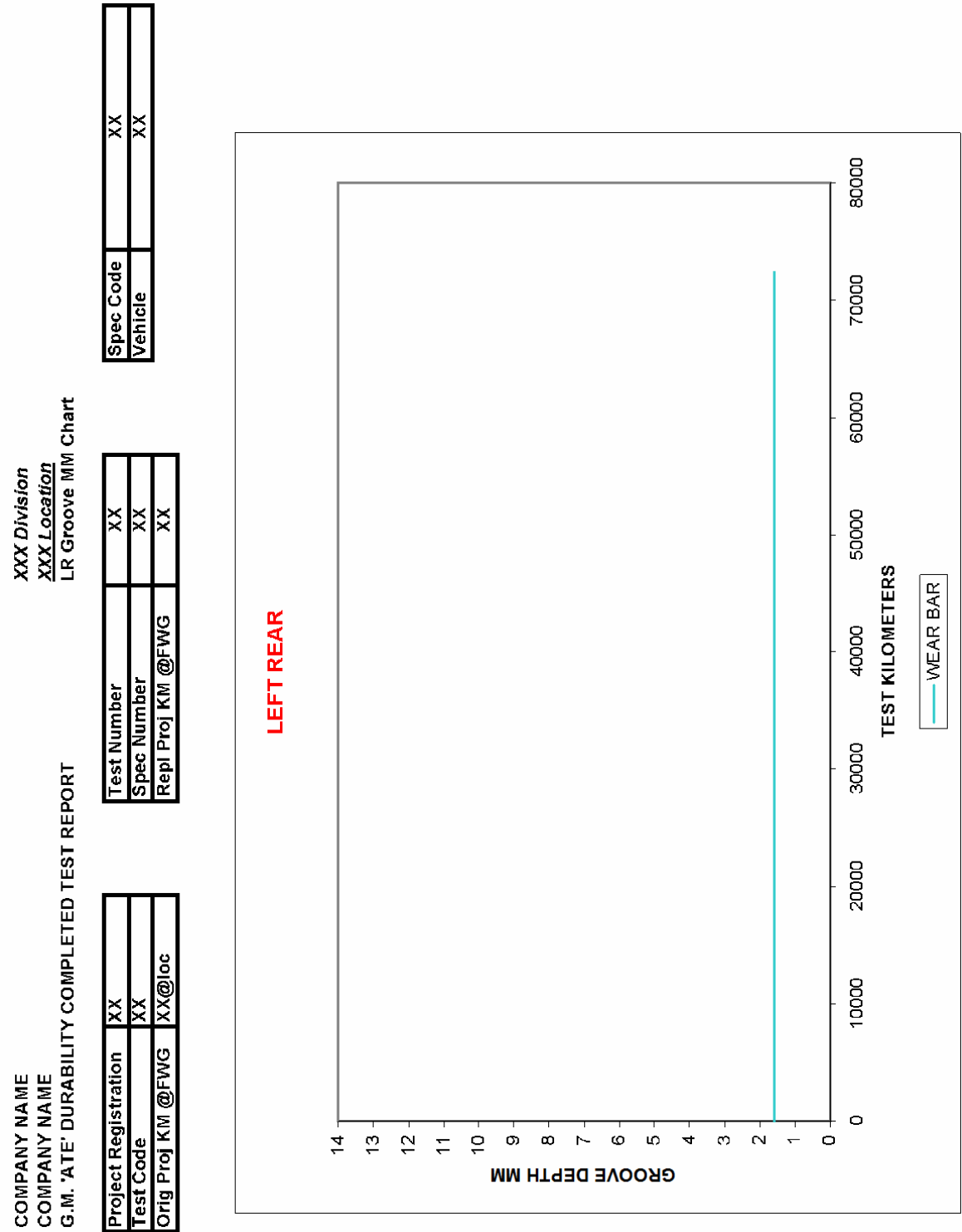
0
8047
16093
24140
32187
40234
48280
56327
64374
72420

AAA
AAA
AAA
AAA
AAA
AAA
AAA
AAA
AAA
AAA

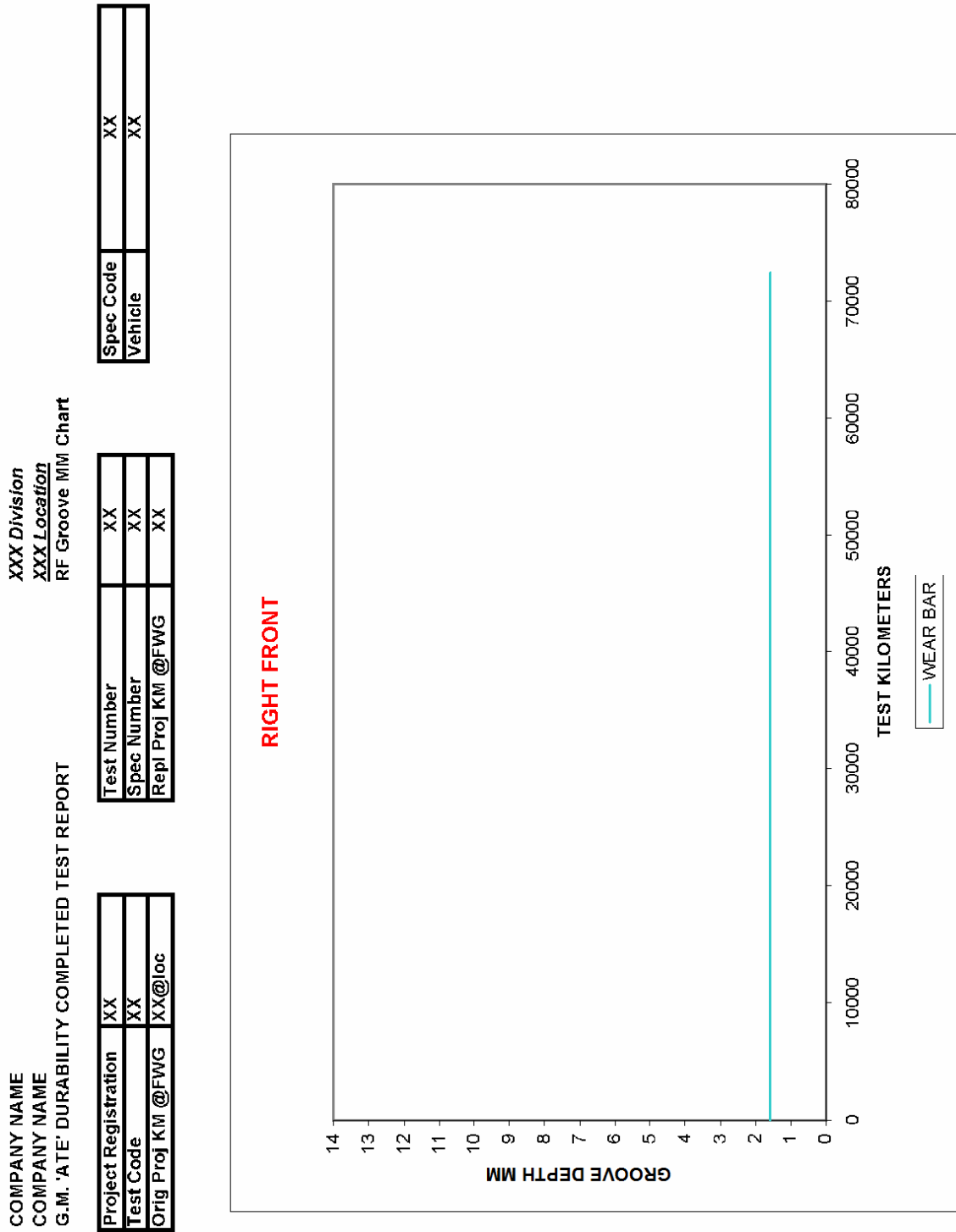
Appendix D - Sample Detailed Test Report Template - Page 11/26



Appendix D - Sample Detailed Test Report Template - Page 12/26



Appendix D - Sample Detailed Test Report Template - Page 13/26



RIGHT FRONT

The graph displays Groove Depth MM on the vertical axis (0 to 14) and TEST KILOMETERS on the horizontal axis (0 to 80,000). A single horizontal line at approximately 1.5 mm is labeled 'WEAR BAR' in the legend.

TEST KILOMETERS	GROOVE DEPTH MM
0	1.5
10000	1.5
20000	1.5
30000	1.5
40000	1.5
50000	1.5
60000	1.5
70000	1.5
80000	1.5

Appendix D - Sample Detailed Test Report Template - Page 14/26

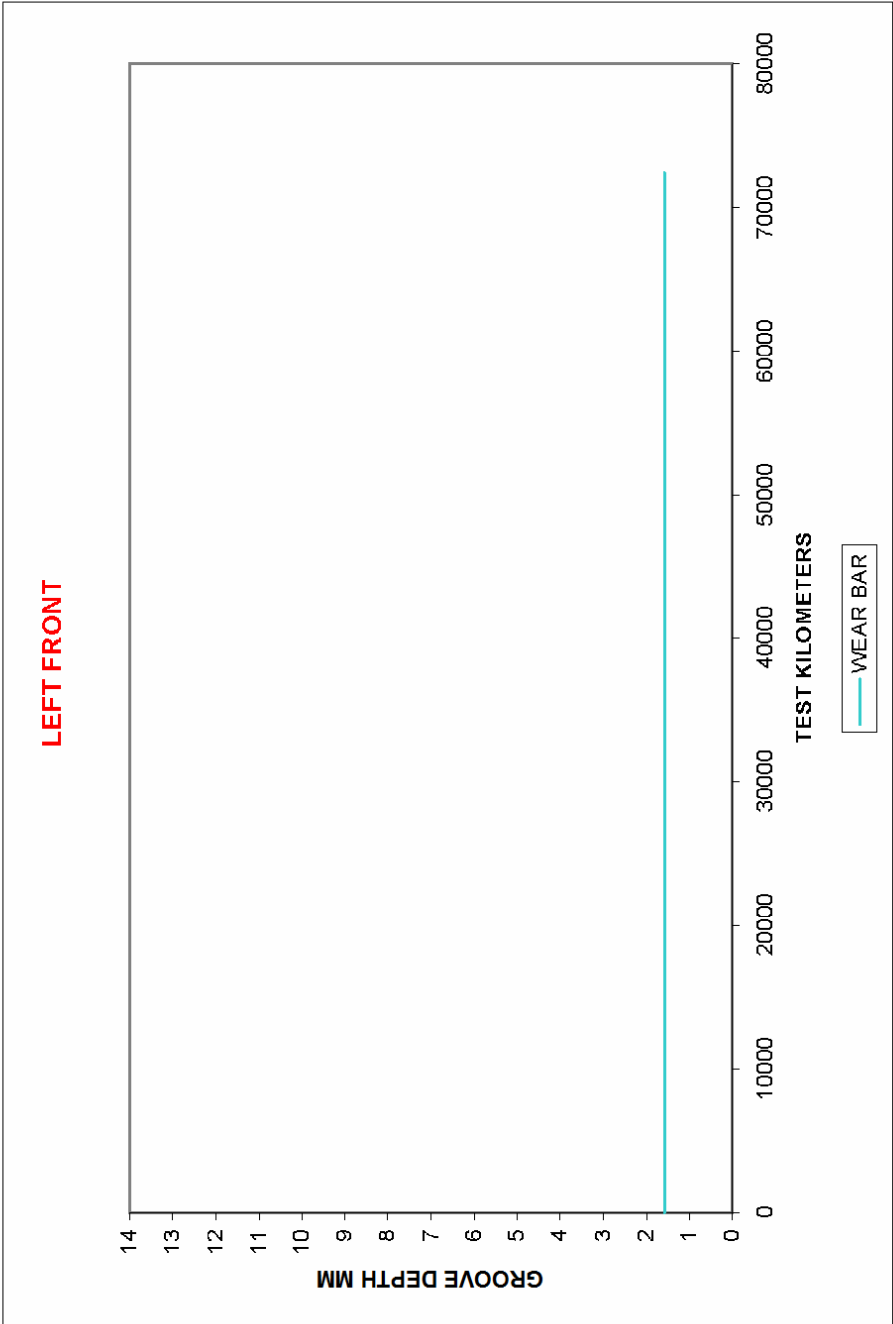
COMPANY NAME
COMPANY NAME
G.M. 'ATE' DURABILITY COMPLETED TEST REPORT

XXX Division
XXX Location
LF Groove MM Chart

Project Registration	XX
Test Code	XX
Orig Proj KM @FWG	XX@loc

Test Number	XX
Spec Number	XX
Repl Proj KM @FWG	XX

Spec Code	XX
Vehicle	XX



Appendix D - Sample Detailed Test Report Template - Page 15/26

COMPANY NAME

COMPANY NAME

G.M. 'ATE' DURABILITY COMPLETED TEST REPORT

XXX Division

XXX Location

Wear Out Progression Report

Test Number

Spec Number

Spec Code

Vehicle

Project Registration

Test Code

Tire Size

XX

XX

XX

0 km

8047 km

16095 km

24140 km

32167 km

40234 km

48280 km

56327 km

64374 km

72420 km

WEAR BAR

LEFT REAR

RIGHT REAR

OSS

OSH

CTR

ISH

ISS

GROOVE

ISS

OSH

CTR

ISH

OSS

Appendix D - Sample Detailed Test Report Template - Page 16/26

COMPANY NAME

COMPANY NAME

G.M. 'ATE' DURABILITY COMPLETED TEST REPORT

XXX Division

XXX Location

Wear Out Progression Report

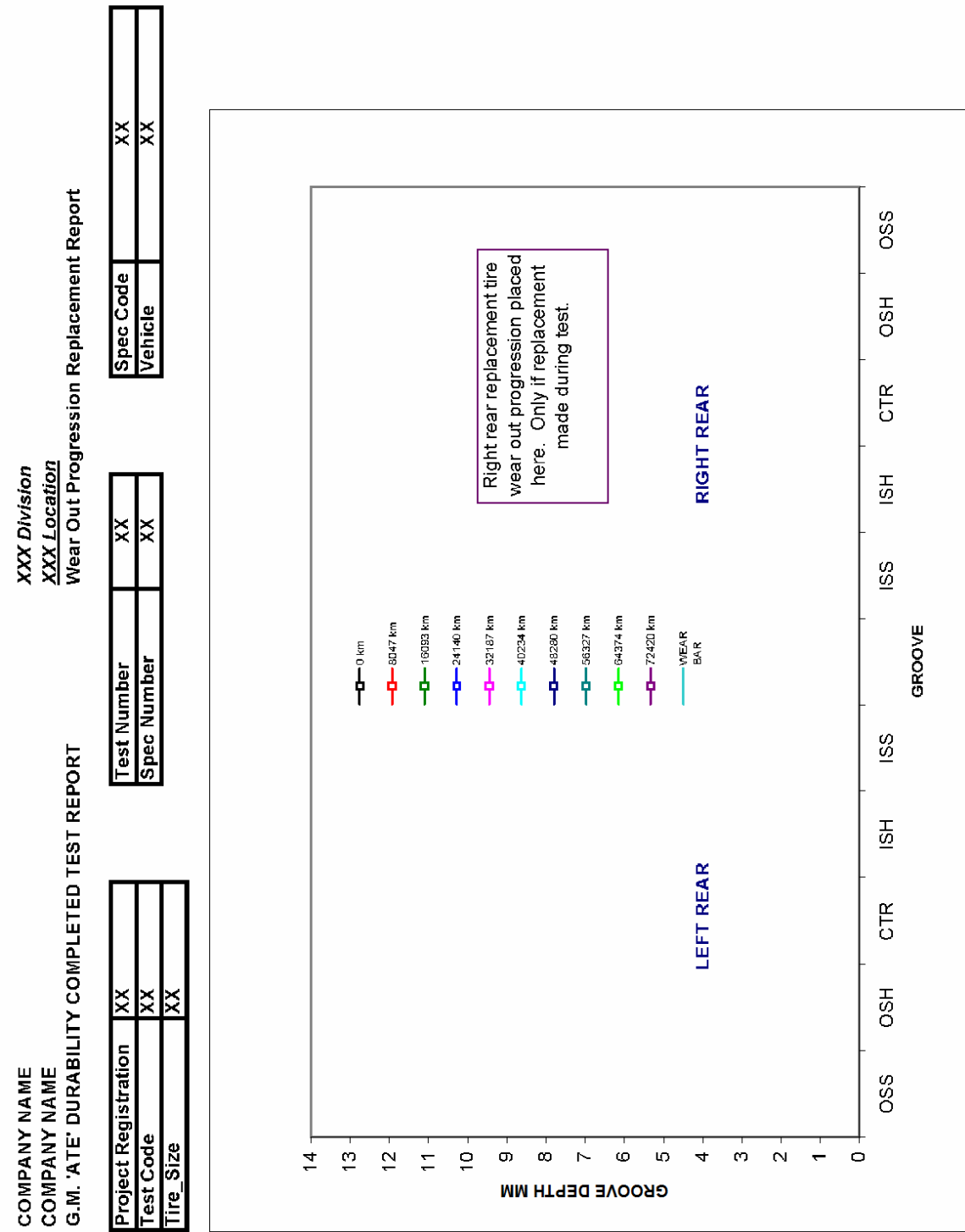
Project Registration	XX
Test Code	XX
Tire_Size	XX

Test Number	XX
Spec Number	XX

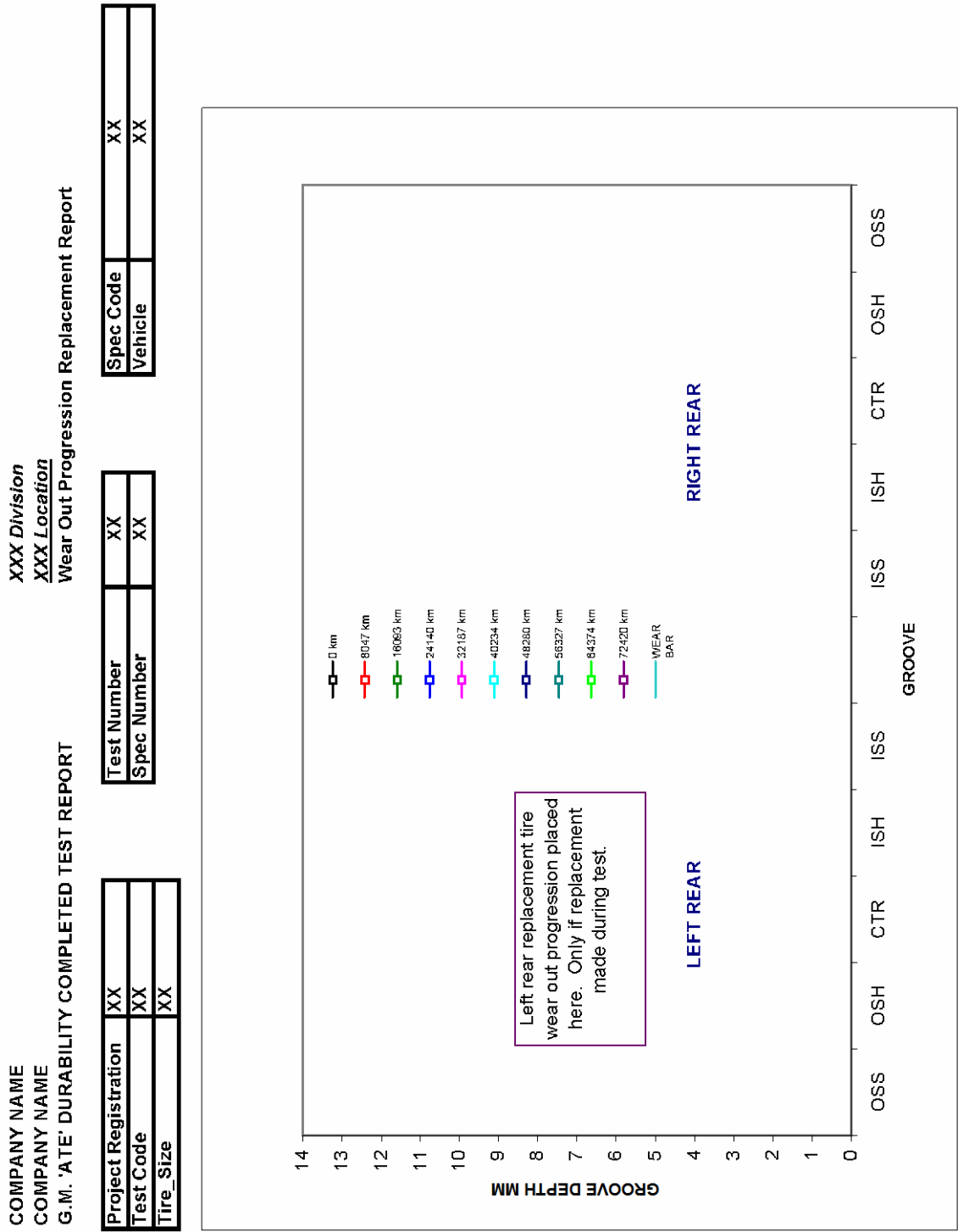
Spec Code	XX
Vehicle	XX



Appendix D - Sample Detailed Test Report Template - Page 17/26



Appendix D - Sample Detailed Test Report Template - Page 18/26



COMPANY NAME

COMPANY NAME

G.M. 'ATE' DURABILITY COMPLETED TEST REPORT

XXX Division

XXX Location

Wear Out Progression Replacement Report

Project Registration

Test Code

XX

XX

Test Number

Spec Number

XX

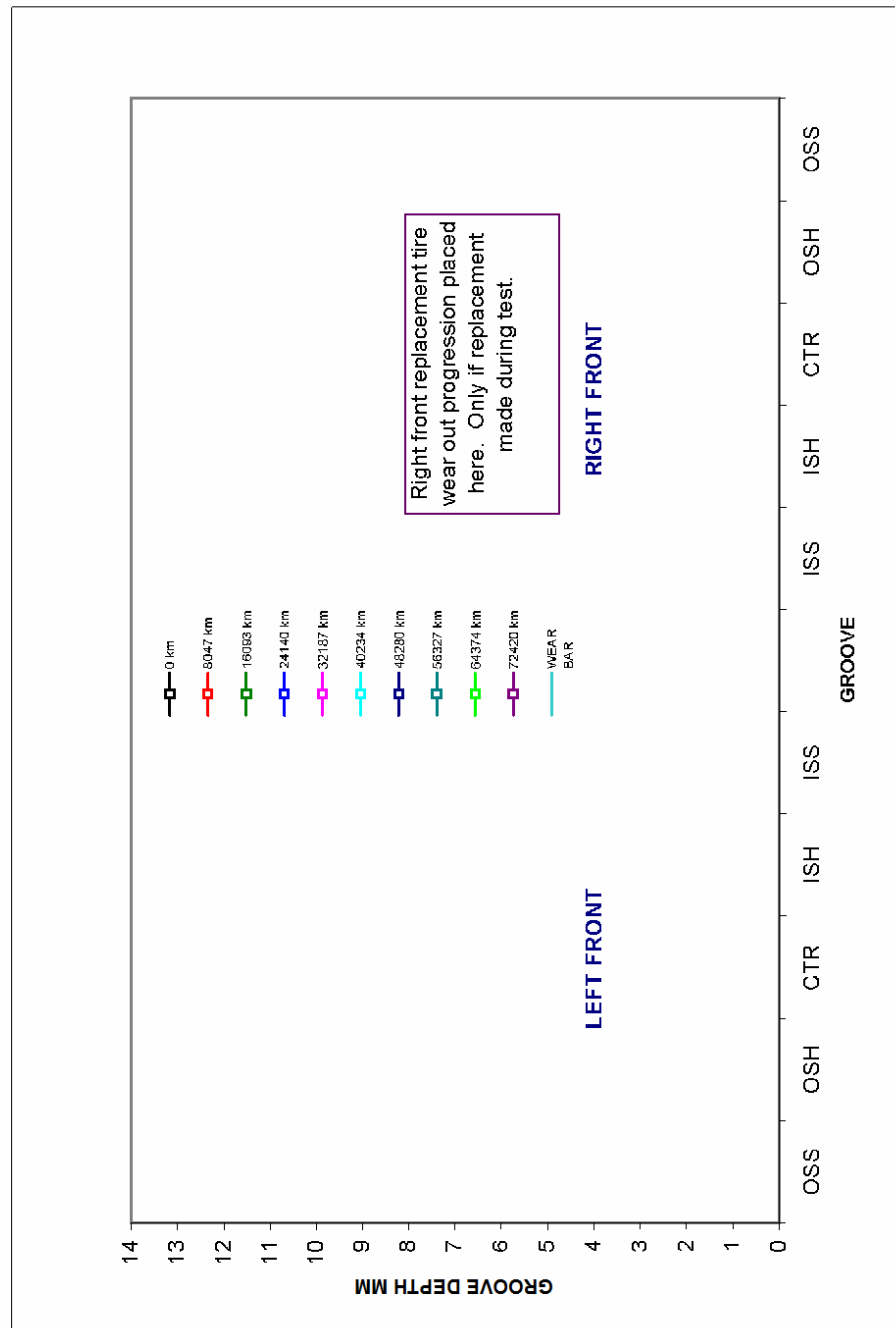
XX

Spec Code

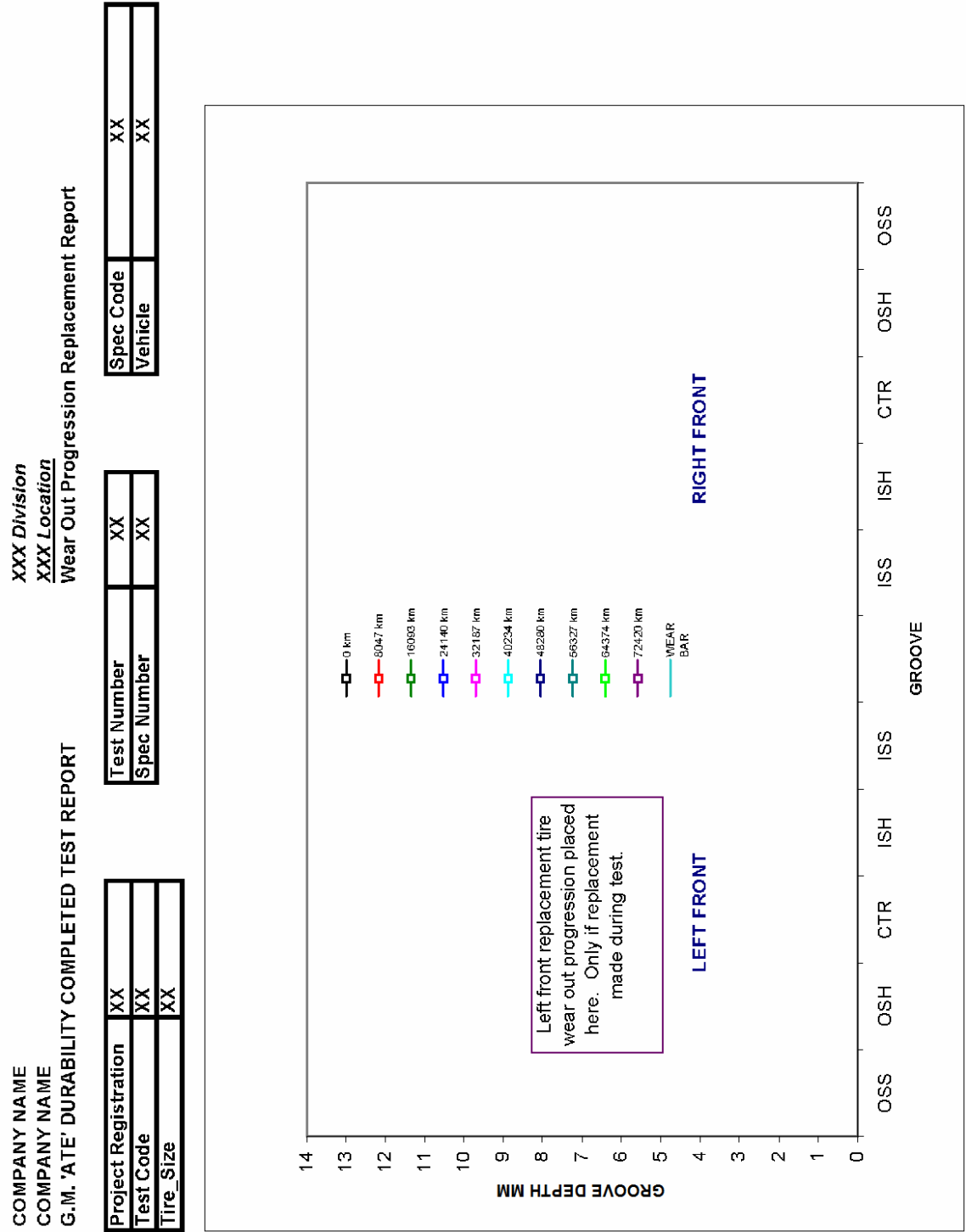
Vehicle

XX

XX



Appendix D - Sample Detailed Test Report Template - Page 20/26



Appendix D - Sample Detailed Test Report Template - Page 21/26

COMPANY NAME

COMPANY NAME

G.M. 'ATE' DURABILITY COMPLETED TEST REPORT

XXX Division

XXX Location

Wear Out Projection Report

Project Registration

Test Code

TireNos

XX

XX

XX/XX

Test Number

Spec Number

XX

XX

Spec Code

Vehicle

XX

XX

100

90

80

70

60

50

40

30

20

10

0

KM TO WEAROUT

OSS

OSH

CTR

ISH

ISS

GROOVE

72420 KW

OSS

OSH

CTR

ISH

ISS

ISH

CTR

OSH

OSS

LEFT REAR

RIGHT REAR

Appendix D - Sample Detailed Test Report Template - Page 22/26

COMPANY NAME

COMPANY NAME

G.M. 'ATE' DURABILITY COMPLETED TEST REPORT

XXX Division

XXX Location

Wear Out Projection Report

Project Registration

Test Code

TireNos

XX

XX

XX/XX

Test Number

Spec Number

XX

XX

Spec Code

Vehicle

XX

XX

100

90

80

70

60

50

40

30

20

10

0

KM TO WEAROUT

LEFT FRONT

RIGHT FRONT

OSS

OSH

CTR

ISH

ISS

GROOVE

72420 KM

ISS

ISH

CTR

OSH

OSS

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Appendix D - Sample Detailed Test Report Template - Page 23/26

COMPANY NAME		Division
COMPANY NAME		Location
G.M. 'ATE' DURABILITY COMPLETED TEST REPORT		
Project Registration Number	XXXX	Test Number
		XXXX

Picture of LF and RF tire treads. Side by side so tread pattern is clearly visible.

Note card in front of tire:

Spec #

On Vehicle Tire location Code

Tire Size

0 miles

Date of picture

Test #

Appendix D - Sample Detailed Test Report Template - Page 24/26

COMPANY NAME COMPANY NAME G.M. 'ATE' DURABILITY COMPLETED TEST REPORT	Division Location	Test Number XXXX
Project Registration Number XXXX	Picture of LF and RF tire treads. Side by side so tread pattern is clearly visible. Note card in front of tire: Spec # On Vehicle Tire location Code Tire Size Final miles Date of picture Test #	

Appendix D - Sample Detailed Test Report Template - Page 25/26

COMPANY NAME		Division
COMPANY NAME		Location
G.M. 'ATE' DURABILITY COMPLETED TEST REPORT		
Project Registration Number	XXXX	Test Number
		XXXX

Picture of LR and RR tire
treads. Side by side so tread
pattern is clearly visible.

Note card in front of tire:

Spec #

On Vehicle Tire location Code

Tire Size

0 miles

Date of picture

Test #

Appendix D - Sample Detailed Test Report Template - Page 26/26

COMPANY NAME COMPANY NAME G.M. 'ATE' DURABILITY COMPLETED TEST REPORT	Division Location	Test Number XXXX
Project Registration Number XXXX	Picture of LR and RR tire treads. Side by side so tread pattern is clearly visible. Note card in front of tire: Spec # On Vehicle Tire location Code Tire Size Final miles Date of picture Test #	